

Subjectivity as an evidential dimension in epistemic modal expressions[☆]

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Abstract

This paper analyzes the role of subjectivity in epistemic modal expressions. It argues for an interpretation of the dimension of subjectivity as a separate evidential qualification. On the basis of corpus data, it attempts to show how the dimension manifests itself in linguistic structure in the context of expression forms of epistemic modality. It reflects on the precise nature of the dimension, and discusses its connections with some other notions of evidentiality. It further considers the relation between the present notion of subjectivity and the notion as found elsewhere in the literature. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

In the literature on epistemic modality, one very often encounters the suggestion that (at least in the West European languages, but probably also in other, typologically divergent, languages) there is a distinction between ‘subjective’ and ‘objective’ evaluations of the likelihood of a state of affairs. The original, and to my knowledge, still the clearest formulation of the distinction is due to Lyons (1977). (But, as Lyons indicates, the concept no doubt relates to the much older distinction in logic between epistemic and ‘alethic’ modality.) Since then, the literature has systematically adopted it, yet this has not resulted in a further clarification of the dimension beyond

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Lyons' definition. On the contrary, though most treatments of epistemic modality somehow mention the dimension of subjectivity (e.g., Coates, 1983; Palmer, 1986), they usually do not go beyond an intuitive characterization of it. It seems fair to say that most authors, while acknowledging the intuitive appeal of Lyons' notion, do not really know what to do with it when it comes to handling concrete empirical data – a fact which underscores Lyons' (1977: 797) own mitigating statement that "its epistemological justification is, to say the least, uncertain".

However, the problem is clearly not only a matter of definition. Another important element is that the literature has so far not offered a clear view of how the dimension relates to linguistic facts: how is it linguistically expressed? Thus, assuming that (for English and other West European languages) the linguistic form types in (1) can express epistemic modality, which of these types are subjective and which are objective? And which of their formal properties mark them as subjective or objective?

(1) a. Modal adverbs:

John probably made it to the bakery before closing time.

b. Modal adjectives:

It is probable that John made it to the bakery before closing time.

c. Mental state predicates:

I think John made it to the bakery before closing time.

d. Modal auxiliaries:

John may have made it to the bakery before closing time.

Lyons himself remains quite vague regarding the connections between the semantic dimension and the linguistic surface (beyond illustrating that the modal auxiliaries can express both subjectivity and objectivity – but then, see Palmer, 1979, and Coates, 1983). Others have attempted to be more specific, yet with quite diverging results (consider quite different analyses in e.g., Perkins, 1983; Kiefer, 1984; Watts, 1984; and Hengeveld, 1988).

In this paper I will argue for an interpretation of the dimension of subjectivity as an evidential qualification which can (but need not) be expressed jointly with the epistemic one (section 2). On the basis of corpus data, I will try to find out how this dimension relates to linguistic expression forms of epistemic modality (section 3). Then I will reflect on the precise nature of the dimension by proposing a (more interaction-oriented) characterization in terms of '(inter)subjectivity', as an alternative to Lyons' definition in terms of sub/objectivity. In support of my reanalysis, I will discuss connections with some other notions in the realm of evidentiality found in the literature (section 4). Finally, I will briefly discuss the relation between the present notion of subjectivity and the notion as it is used elsewhere in the literature (section 5).

Before embarking, however, two preliminary remarks are in order. First of all, the concept of 'subjectivity' under consideration here (and in the literature cited) should not be confused with the notion of 'speaker commitment' (i.e., involvement of the speaking 'subject'). All the examples in (1) involve the speaker's commitment to the epistemic expression. That is, in each, the use of the modal expression implies that

at ‘utterance time’ the speaker subscribes to and accepts responsibility for the epistemic evaluation underlying it. Or, to use yet other terminology, in each of these examples the modal expression is used ‘performatively’. This contrasts with ‘descriptive’ uses, such as *Mary thinks that John made it to the bakery* or *Mary considers it probable that John made it to the bakery*, in which the speaker reports on someone else’s epistemic evaluation of a state of affairs without there being any explicit indication as to whether the speaker personally subscribes (i.e., is committed) to the veracity of the evaluation or not (see also Nuyts (2000b) on performativity vs. descriptivity in epistemic modal expressions). The present issue of subjectivity is orthogonal to the matter of performativity vs. descriptivity; hence, it applies within the range of performative uses of epistemic expressions.

Second, a brief note on terminology. In this paper, I will use the label ‘dimension of subjectivity’ as a shorthand expression whenever I need to refer to the dimension as a whole, covering the two poles involved, viz. subjectivity and its counterpart, objectivity or intersubjectivity (see section 4). The label is conveniently general whenever the precise construal of the ‘meaning’ of the dimension is not relevant. As soon as that construal *does* matter, however, I will switch to more specific labels.

2. Subjectivity as an evidential dimension

Let us start by taking a closer look at Lyons’ (1977: 797ff.) definition of the dimension of subjectivity. According to him, while objective epistemic modality expresses an objectively measurable chance that the state of affairs under consideration is true or not, subjective epistemic modality involves a purely subjective guess regarding its truth. Consider (2):

(2) Alfred may be unmarried.

In Lyons’ view, (2) may either indicate that the speaker is simply uncertain about the (hypothetical) fact that Alfred is unmarried – i.e., subjective modality. Or, alternatively, it may indicate that according to the speaker there is a mathematically computable chance that Alfred is unmarried, for example because (the speaker knows that) Alfred belongs to a community of ninety people, of which thirty are unmarried, hence there is one chance in three that he is unmarried – i.e., objective modality.

As suggested above, there is a strong intuitive appeal to this concept, but the issue should probably be approached differently than it is in current practice. A first element of this concerns the status of the dimension. Most authors (including Lyons) appear to interpret the dimension as being inherent to epistemic modality as such, i.e., as marking a difference between two ‘types’ of epistemic modality (cf. e.g., Hengeveld, 1988 for a very explicit statement to this effect). Yet, there is reason to believe that this is not exactly the right way of looking at it. A conceptual analysis of Lyons’ characterization of the two readings of (2) may pave the way for an alternative (and, as I will show later, empirically more viable) view.

What does ‘being simply uncertain about a fact’ (i.e., the presumed subjective meaning) mean? Any statement as to the likelihood of a person being married or not is at least based (consciously or unconsciously) on the knowledge that in our society some people are married and others are not, hence there is always a chance that Alfred is (un)married. Depending on how well one knows Alfred, there may be other factors, such as knowledge about his age, his character (could there be anyone willing to live with him?), his lifestyle (is he always accompanied by the same woman or by different ones, or is he always alone?), etc., which may codetermine one’s judgment of the chances that Alfred is (un)married. So, in this case, just as in the situation in which one knows that Alfred belongs to a community of ninety people of which thirty are unmarried, the judgment is, no doubt, based on some kind of evidence (in a wide sense, including directly perceived evidence, hearsay evidence, inferences from other knowledge, etc.). In principle, if one does not have any kind of evidence pertaining to a state of affairs, one cannot evaluate its probability. In such circumstances one can only say that ‘one does not know’. Surely, people often express epistemic judgments which are not actually based on reasonable evidence at all. But by expressing the judgment they are nevertheless suggesting (or rather feigning) that they do have evidence for their statement. This is a matter of (in)sincerity: they are then misleading the listener.

Of course, what may vary is the quality and/or nature of the evidence one has for an epistemic judgment. Thus, if for the time being we accept Lyons’ formulation, one could reframe the distinction (vaguely) in terms of how ‘good’ the evidence is for an evaluation: an epistemic evaluation based on better (more reliable) evidence would probably be experienced as being ‘objective’, while one based on shaky evidence would rather be considered more ‘subjective’. (As suggested in the introduction, and as will be elaborated below, this is not the only possible way, and certainly not necessarily the most appropriate way to construe the ‘meaning’ of the dimension of subjectivity in epistemic expressions. However, this does not affect the present point.)

Now, note that this analysis of the matter involves two different (though related) semantic dimensions. On the one hand, it involves the speaker’s evaluation of the probability of the state of affairs, i.e., the epistemic qualification. On the other hand, it involves the speaker’s evaluation of the quality of the evidence for that qualification, i.e., a dimension which would rather go under the umbrella of what would traditionally be labeled evidentiality (but see section 4).¹ The dimension of subjectivity is thus probably not a distinction within the epistemic domain, but within the evidential domain. And if so, we are not dealing with two types of epistemic modality, but with an interaction of an epistemic with an evidential qualification.

¹ It is, of course, not one of the ‘classical’ evidential notions, such as ‘inferential’, ‘hearsay’, or ‘experiential’, which all mark the type of source for one’s knowledge about a state of affairs. But to the extent that it concerns elements/properties of one’s evidence for the information provided in the utterance, it does in a way fit the definition of the global category of evidentiality.

There are at least two further arguments in direct support of this assumption, viz.:

- (i) The dimension of subjectivity is probably not limited to epistemic modality. Lyons (1977) argues that it also applies to deontic modality. And its range appears to go even further than that: at least in English and other West European languages, there are several independent expression forms for subjectivity (as one of the two poles of the dimension). Thus, expressions such as *in my view/opinion/mind, if you ask me, to me*, etc., can be used to indicate that the speaker is offering information about a state of affairs which is strictly subjective, even in the absence of any other (epistemic, deontic, or whatever) qualificational expression in the utterance (e.g., *if you ask me, John did not go to the bakery at all*). (If the other pole of the distinction is construed as ‘intersubjectivity’ – see below – then there appears to be independent terminology for it, too: e.g., *it is known that ...*) If this is adequate, it obviously strengthens the analysis in the above terms, and disfavors the ‘classical’ analysis: If the dimension of subjectivity also occurs jointly with other qualificational dimensions, and even as an ‘independent’ qualification of a state of affairs, then it is clearly more parsimonious to assign it the status of a separate semantic dimension. The precise extension of the dimension is an interesting venue for future research, but is beyond the scope of the present paper.
- (ii) If, as the ‘classical’ analysis presumes, the dimension of subjectivity were inherent to epistemic modality, then one would expect that all epistemic expression forms would feature to one of the two values involved in it. This, however, is not the case, as will be shown in section 3.

In addition, there is a third element which supports the above reanalysis, at least indirectly:

- (iii) If (unlike Lyons’ original characterization) the dimension is defined in terms of ‘(inter)subjectivity’, then it appears to be related to some other evidential dimensions reported in the literature to occur in a number of typologically divergent languages. I will turn to this in section 4.

3. The linguistic reflection of the dimension of subjectivity in epistemic expressions

As mentioned in the introduction, so far the literature has not offered a clear and convincing analysis of the link between the dimension of subjectivity and linguistic expressions of epistemic modality. Part of the problem is no doubt that authors – due to their conception of the issue as inherent to the dimension of epistemic modality itself – have been forced to look for this link even in places where it is not to be found. Let us see, then, what we can learn from an analysis of instances of the epistemic expression types listed in (1) above as they occur in corpus data.

The data for the following are drawn from a large corpus of spoken and written Dutch and German. For Dutch, I have used the Uit den Boogaart (1975) corpus and a corpus compiled by Luc Vandenbosch (1992). For German, I have used a selection from the large system of corpora available at the ‘Institut für deutsche Sprache’ in Mannheim (online, conditional upon written agreement), and specifically from the ‘Mannheimer Korpus 1’, the ‘Handbuchkorpus 1987’, the ‘Grammatikkorpus’, and the ‘Dialogstrukturenkorpus’. The written data involve expository prose from journals, magazines, and popular scientific publications (more than 500,000 words for Dutch and German each), and literary and pseudoliterary prose from novels (more than 150,000 words per language). The spoken data (more than 200,000 words per language) are (predominantly) transcriptions of TV and radio interviews and debates, and (to a lesser extent and only for German) of counseling sessions and university oral examinations. In these corpora, I have focused on the occurrences of one specific (hopefully representative or prototypical) linguistic form per expression type listed in (1) (see section 1), and per language, viz. the Dutch resp. German adverb and adjective *waarschijnlijk* and *wahrscheinlich* (‘probably/probable’),² the mental state predicate *denken* (‘think’) in Dutch, and *glauben* (‘believe’) in German,³ and the Dutch resp. German modal auxiliary *kunnen* and *können* (‘may’).⁴ In the examples below, the relevant forms are italicized.⁵

The following is but a very synoptic overview of the results of the full corpus analysis. It focuses on the global patterns emerging across the different discourse types represented in the data (with attention paid to differences between discourse types only when they are crucial for understanding the global pattern). This overview will not provide quantitative information. A more detailed presentation and discussion of the data is offered in Nuyts (2000b).

In order not to anticipate and/or cause confusion in view of the discussion of the nature of the dimension of subjectivity in section 4, the two poles will provisionally be labeled ‘subjectivity’ and ‘non-subjectivity’ (rather than ‘objectivity’). ‘Non-subjectivity’ should, thus, not be conflated with ‘neutrality’. The criteria used to characterize instances in the data as either subjective or non-subjective have been widely set so as to include any more or less systematically construable interpretation of the opposition (possible interpretations are discussed in section 4).

² Unlike English, neither Dutch nor German have morphological marking of the difference between adverbial and adjectival uses of this form.

³ In Dutch and English, *denken* and *think* are by far the most frequently used (epistemic) qualificational mental state predicates (see, e.g., Thompson and Mulac, 1991, for frequency data on English). However, German *denken* is hardly ever used this way: it occurs nearly exclusively in its ‘literal’ mental process reading, i.e., as referring to the mental act of thinking. Instead, *glauben* is the most common form for epistemic uses.

⁴ For the mental state predicate and the auxiliary, I have used only part of the corpus mentioned above, in order to keep the number of occurrences within manageable limits. See Nuyts (2000b) for details.

⁵ For the sake of the presentation, the corpus examples have been ‘cleaned up’ in a few minor respects: irrelevant pauses, hesitations, repetitions, repairs, back channel cues, etc. are omitted. Spoken examples have no initial capitalisation, not even in German; (in line with the conventions in the corpora used) interpunction in them more or less renders the Speaker’s prosody.

3.1. Modal adverbs

The modal adverbs are essentially neutral in terms of the dimension of subjectivity. In specific contexts they may involve a subjective or a non-subjective evaluation. Thus, most authors (Kiefer, 1984; Watts, 1984; Hengeveld, 1988) have maintained that adverbs express subjectivity, and, true, they do occur in circumstances in which they can be construed as involving a subjective evaluation. But on the other hand, in occurrences such as (3) or (4), the author is reporting on the results of long term research performed by a (more or less) large community of scientists, and his/her opinion can hardly be considered a matter of a purely subjective commitment.

- (3) De oudste cultuurlagen in Japan tonen ons een jagers- en vissersvolk, zeer *waarschijnlijk* afstammelingen van immigranten van het vasteland van Azië.
'The oldest cultural layers in Japan show us a people of hunters and fishermen, very probably descendants of immigrants from the Asian mainland.'
- (4) Alle Sterne in einem solchen Sternhaufen sind sehr *wahrscheinlich* etwa gleichzeitig aus einer gemeinsamen großen Gaswolke entstanden.
'All stars in such a cluster very probably developed at about the same time out of a single large cloud of gas.'

Moreover, most occurrences of the adverbs in the corpus data appear not to suggest anything at all in terms of the dimension of subjectivity – they are perfectly neutral.⁶ To the extent that the adverbs do show an effect in the range of the dimension of subjectivity, then, this is clearly not due to the adverbial form, but is purely contextual.

3.2. Modal adjectives

The dimension of subjectivity is systematically involved in adjectival expressions of epistemic modality. In fact, the adjectival construction can express either of the two poles, depending on the form of the construction. It expresses non-subjectivity in the 'standard' form of the type *it is probable that*, involving an impersonal subject and the use of the copula *be* which "categorically asserts the modality expressed" (Perkins, 1983: 67). Hence this pattern occurs by far most frequently in circumstances which 'match' its non-subjective nature, i.e., in reports on the results of scientific research, as in (5), and/or in contexts involving explicit 'logical' reasoning, as in (6) (the latter is often linked to scientific contexts, but there are also instances involving common sense reasoning in non-scientific discourse).

- (5) In der archaischen Zeit hatte die Mauer bereits denselben beträchtlichen Umfang wie in späterer Zeit. Es ist sehr *wahrscheinlich*, daß dieser älteste

⁶ Recall – see section 1 – that this does not mean that there is no speaker commitment involved in the use of these adverbs. On the contrary, adverbs are always and unavoidably 'performative' – see Nuyts (1993, 2000b).

Mauerring, der für die damaligen Verhältnisse und Vorstellungen ungewöhnlich groß war, von dem Tyrannen Polykrates errichtet wurde, [...].

'In the archaic era the wall already had the same considerable circumference as in later times. It is very probable that this oldest circular wall, which was unusually large for the standards and conceptions of those days, was erected by the tyrant Polykrates.'

- (6) Chauvin zette de proeven voort. Hij plaatste bijvoorbeeld een groene larve in een afgesloten fles tussen een hele partij zwart-rode soortgenoten. Prompt werd die larve ook zwart-rood. Maar zag het minuscule ding dan de anderen en paste het vervolgens zijn kleur bij die van de anderen aan? *Zeer onwaarschijnlijk!*
'Chauvin continued the experiments. For example, he put a green larva in a closed jar between a whole set of black-red individuals of the same kind. Instantly this larva also turned black and red. But did the tiny thing see the others, then, and did it consequently adapt its color to that of the others? Highly unlikely.'

In variants of the pattern in which the speaking subject enters the scene, however, it immediately turns subjective, as is shown in (7) (the double negation further adds to the subjectification here) and (8).

- (7) Mijn ontstemming werd nog groter toen mijn kameraden even later met een zelfvoldane gelaatsuitdrukking terugkeerden, al leek het mij niet *onwaarschijnlijk* dat zij maar branieachtig deden alsof zij een opwindend avontuur achter de rug hadden.

'My annoyance increased even more when shortly afterwards my friends returned with self-satisfied faces, even though it seemed not unlikely to me that they were only bragging as if had an exciting adventure behind them.'

- (8) Ich halte derzeit aber ein reales Wachstum von gut einem Prozent noch für *wahrscheinlich*.

'But I presently consider a real growth of at least one percent still probable.'

Such subjective uses are rare for Dutch *waarschijnlijk* and German *wahrscheinlich* ('probable'/'likely'), but for adjectival expressions of epistemic certainty there are even strongly idiomatic constructions in West Germanic: cf. English *I am sure (of it) (that ...)*, Dutch *ik ben (er) zeker (van) (dat ...)*, German *ich bin mir sicher (daß ...)*. At least in Dutch, a quick survey of the corpus indicates that this form occurs as frequently as the impersonal (hence non-subjective) variant.

3.3. Mental state predicates

The mental state predicates systematically express subjectivity.⁷ Thus, they typically and predominantly occur in contexts in which the speaker voices personal opinions, very often about topics in the realm of strictly individual experiences or

⁷ At least, I have evidence that English *think* and *believe*, Dutch *denken* and *geloven*, and German *glauben* do. But presumably the same is true for other mental state predicates, such as English *suppose*

concerns, or also in contexts involving antagonism between the views of speaker and hearer. Consider for example an interview in the Dutch spoken data involving 5 Catholic clergymen/women who were asked why they have taken the vow and how they think of their (many years of) service in the church, i.e., an interaction which abounds in strictly personal remarks about very subjective feelings. The number of occurrences of *denken* in this interview is nearly triple the average in spoken Dutch in general, and 10 times the average for spoken and written Dutch altogether. For a comparison: the number of adverbs in this interview perfectly matches the average for spoken Dutch, and there are no adjectives in it. Another (if less dramatic) example involves a political debate in the German spoken data, in the aftermath of a large demonstration against a planned nuclear power plant, i.e., a highly antagonistic type of interaction. The frequency of *glauben* in it is double the average for spoken German, or nearly 5 times the average for spoken and written German altogether. In contrast, there are no adjectives in the transcript of this debate, and even the frequency of adverbs is well below the average for spoken German. Examples (9) and (10) illustrate these interactions:

- (9) maar ik *denk* dat ook in het geloof up en downs zijn, want geloof heeft voor mij alles te zien met het leven en omgekeerd.
 ‘but I think that also in believing there are ups and downs, because to me believing has everything to do with life and vice-versa.’
- (10) da widersprech ich jetzt ganz grundsätzlich. sie könnten noch tausend veranstaltungen machen mit denen die dagegen sind, sie würden kein jota daran ändern daß sie sagen bei uns nicht. ich *glaube* das müssen sie sehn.
 ‘Now that I contradict very fundamentally. You could organize a thousand meetings with those who are against it. You wouldn’t at all change their saying “not here”. I think you have to see that.’

Because the mental state predicates are inherently subjective, they are frequently used as mitigating or hedging devices, as illustrated in (11) and (12). In such uses, it is usually quite obvious that speakers are absolutely certain about or convinced of what they are saying, but by using the mental state predicate they suggest that they are voicing a tentative and personal opinion which may be wrong, thus ‘officially’ leaving room for another opinion or for a reaction on the part of the hearer.

- (11) wel eh- ik *dacht* dat ik dat al gezegd had zoeven, hé.
 ‘Well, I thought that I had already said that a minute ago, didn’t I?’
- (12) ich *glaube* jetzt muß ich doch etwas sagen herr kollege.
 ‘I think now I have to say something after all worthy colleague.’

or *guess* and German and Dutch equivalents, although these other predicates appear to express other evidential meanings in addition. However, unlike *think*, *believe* and their Dutch and German equivalents, there are reasons to assume that these other mental state predicates no longer express epistemic modality – at least not directly – i.e., they are much rather subjective evidential markers. See Nuyts (2000b) for discussion.

3.4. Modal auxiliaries

The modal auxiliary is neutral in terms of the dimension of subjectivity – as can be inferred from Lyons' illustration of the dimension by embedding a single sentence with a modal auxiliary in two different contexts. The modal auxiliary only receives a (non-)subjective interpretation when the context imposes this – that is, when the meaning of the modal auxiliary unavoidably 'absorbs' the flavor of those contexts. But most of the occurrences appear perfectly neutral in these terms, exactly like the adverbs.

This is true, at least, for the normal, 'sentence internal', dependent use of the modal, as in (1d) above. Maybe an exception should be made for one special type of use, viz. in a complementing pattern of the kind in (13).

(13) It may be that John has made it to the bakery before closing time.

This pattern is very rare in the Dutch and German data (it has probably developed only very recently, at least in Dutch), so conclusions about it are highly tentative. Still, although this pattern occurs neither in scientific (i.e., clearly non-subjective) nor in clearly subjective contexts, in several occurrences it does serve to formulate a (more or less) 'logical' conclusion drawn from evidence or common sense arguments explicitly introduced in the context (hence 'publicly available'). Examples are (14) (the evidence is the bank withdrawals) and (15) (the arguments for the conclusion have been introduced at length in the preceding context but they are too long to be reproduced here).

(14) vanaf 18 mei hebben wij geen enkel teken van leven niet meer van mijn zoon gevonden. het zou *kunnen* zijn dat wij van 13 tot 18 mei een teken van leven hebben indien dat hij bankopvragingen heeft gedaan.

'After May 18th, we have not found any signs of life from my son anymore. It might be that from May 13th till 18th we do have a sign of life if he has made any bank withdrawals.'

(15) nur eben die gewisse skepsis die ich eben auch habe daß es also [...] sehr wohl sein *kann* daß diese filme [...] en sehr schädlichen effekt haben.

'just only this certain skepticism which I also have that it indeed may be that these movies have a very harmful effect.'

In view of such cases, one might tentatively assume that this specific pattern is acquiring a non-subjective meaning diachronically – which would not be surprising in view of the impersonal subject and a syntactic structure not unlike that of the adjectives.

In summary, (a volume of) the dimension of subjectivity inherently expressed by some epistemic expressions, but not by others. As indicated in section 2, this implies that it can hardly be a dimension inherent to the epistemic evaluation as such, for then one would expect it to be always present. Also note that the subjectivity or non-subjectivity effect, when it occurs, is always due to the syntactic structure 'selected'

by the epistemic lexical item, viz. an impersonal subject and corresponding syntax in the ‘non-subjective’ adjectives and complementing auxiliaries, and a directly related first person subject or argument in the ‘subjective’ adjectives and the mental state predicates. The effect is, thus, not due to the lexical item itself (cf. also the fact that there are subjective and non-subjective variants of the adjective, and neutral and non-subjective variants of the modal, even if in both cases the lexical item expressing the epistemic evaluation is identical).⁸ This, too, pleads for treating this dimension as something separate from the epistemic evaluation.

In the literature, there have been attempts to relate a series of semantic-syntactic ‘behavioral properties’ of (some of) the epistemic expression types to the dimension of subjectivity, such as their interaction with negation and tense or their behavior in conditionals and questions (e.g., Lyons, 1977: 799; Perkins, 1983; Hengeveld, 1988). Going into these would exceed the space allotted to the present paper. As argued in Nuyts (1992, 1994, 2000b), however, these behavioral features can be better explained in terms of other functional properties of the expression types, such as their information structural characteristics, or their ability to be used not only performatively but also descriptively. Accordingly, these behavioral properties have little or nothing to do with the issue of subjectivity.

4. How to define the dimension of subjectivity

As suggested above, Lyons’ (1977) characterization of the dimension of subjectivity is not the only possible one, and is not necessarily the most obvious one to account for the facts of at least these West Germanic languages. Purely intuitively, one can construe the opposition in at least two ways.

- (i) As discussed in section 2, in (my reassessment of) Lyons’ view, the dimension concerns the quality of the speaker’s evidence for an epistemic evaluation: does (s)he have good, mathematically or formally reliable evidence (i.e., objectivity), or does (s)he have poor or vague, intuitive evidence (i.e., subjectivity).
- (ii) Alternatively, the dimension can be defined in terms of a difference in the status of the evidence and the epistemic evaluation based on that evidence from the perspective of the (knowledge of the) interaction partners in that situation. The alternatives within this dimension could then be phrased as follows: one pole involves the speaker’s indication that (s)he alone knows (or has access to) the evidence and draws conclusions from it; the other pole involves his/her indication that the evidence is known to (or accessible by) a larger group of people who share the same conclusion based on it. In the former case the speaker assumes strictly personal responsibility for the epistemic qualification, in the latter case (s)he assumes a shared responsibility among those who have access to the evidence and accept the conclusions from it (including him/herself). An

⁸ There are reasons to believe the subjective meaning element *has* entered the lexical meaning of the mental state predicate itself. See Nuyts (2000b) for arguments.

important (but maybe not a necessary) element in how this distinction actually works may then be whether the interaction partner belongs to those sharing the evidence and the conclusions, or not. In fact, it is important to realize that in any construal of the dimension, its reflection in actual expressions of epistemic modality is a matter of how the speaker presents the situation (rather than of how it 'really is'). In the present understanding of the dimension, then, the position of the interaction partner is likely to become critical.

To label the poles in the first construal I will use the classical terms 'subjectivity' vs. 'objectivity' (with the understanding that the status of this dimension as analyzed here differs from the 'classical' analysis, as discussed in section 2). For the poles in the second construal I will use the terms 'subjectivity' vs. 'intersubjectivity'.

In a way, both construals have an intuitive appeal. Both may play a role in verbal behavior in general, and in issuing epistemic judgments in particular; they are not necessarily mutually incompatible. That is, it is not to be excluded that they are somehow both relevant in determining the choice of epistemic expression types. Yet, there are at least two arguments suggesting that the (inter)subjectivity reading may be closer to what is really involved in the 'intuition' behind the dimension of subjectivity in epistemic expressions. They are the following:

4.1. (Inter)subjectivity better matches the corpus data

Looking back upon our discussion of the corpus data in section 3, the (inter)subjectivity reading appears to have a (slight) preference when it comes to accounting for the observations. In many contexts involving what was called the 'non-subjective meaning' in section 3, it is possible to construe the meaning of the dimension either way. For example, epistemic evaluations occurring in the context of popular scientific reporting, as in (3), (4), or (5), can be construed as being based on solid evidence (i.e., 'objective'). But they can just as well be construed as being generally accepted in the scientific community and by those interested in its achievements (such as readers of popular scientific works), i.e., as being shared by a large group of people, including the author, and from the moment of reading onward presumably also the reader (i.e., 'intersubjective'). In some contexts, however, construing an 'objective' reading appears difficult. For example, in cases in which the epistemic evaluation is based on premises made explicit in the surrounding discourse context, construal in terms of the 'quality' of the evidence ('objectivity') is less sensible than construal in terms of shared knowledge (due to the explicit mentioning of the evidence/arguments) of the reasoning leading to the epistemic evaluation ('intersubjectivity'). This is clear in cases such as (6), (14), or (15), and especially in the cases involving common sense reasoning.

The cases labeled 'subjective' in section 3 are best construable in terms of the (inter)subjectivity dimension rather than in terms of the sub/objectivity dimension, too. For example, in contexts of strictly personal experiences, as in (7) or (9), the quality of the evidence probably matters less than the fact that it is evidence only available to the speaker, not to the hearer. In (9), e.g., the evidence is probably direct

experience, which would count as the best possible quality, yet the critical point is that it is personal experience, not shared experience. Even more clearly, cases which can be characterized as antithetical, such as (8) and (10), as well as mitigating or hedging uses, such as (11) and (12), fit the definition of the dimension in terms of (inter)subjectivity much better than the definition in terms of quality.

On the other hand, in my data (at least) I have not encountered cases in which the sub/objectivity reading is clearly preferred over the (inter)subjectivity interpretation. If one (reasonably) assumes that it is always the same dimension of subjectivity that is involved in the different uses of epistemic expressions, then these observations clearly point in the direction of (inter)subjectivity as the relevant dimension.

4.2. *(Inter)subjectivity has typological correlates*

I am not aware of literature pointing out any role for sub/objectivity (i.e., the marking of the quality of evidence) in evidential marking in languages of the world. But the (inter)subjectivity reading *does* remind one of a few evidential or evidential-like dimensions observed in typologically quite divergent languages. So interpreting the dimension of subjectivity (in epistemic expressions and in general) in West European languages in terms of (inter)subjectivity brings it in tune with a semantic category of wider typological significance.

Thus, there is a very obvious link with the opposition between the ‘habitual experiential’ and ‘gnomic’ categories of evidential marking in Sherpa observed by Woodbury (1986): the former indicates that the speaker has direct experiential evidence, while the latter indicates that the speaker states something which (s)he has not directly experienced him/herself but which is generally known.

Likewise, there is a clear similarity to the evidential meaning of the ‘informal ending -*ô*’ in Korean as discussed in Lee (1993): this form is used for information over which the speaker has exclusive authority, as opposed to other markers which indicate kinds of evidentiality for which the speaker is not solely responsible.

It is also not farfetched to see a link with the (quite broad) category called ‘mirativity’ (DeLancey, 1997), i.e., the marking of information as new or surprising.⁹ To see the relation more clearly, let me briefly elaborate on the definition of (inter)subjectivity given above. In this construal of the dimension, subjectivity is probably very often a matter of formulating a hypothesis ‘on the spot’, without having thought much about it beforehand (hence it is strictly personal, but also potentially new

⁹ This notion of ‘newness’ – as discussed in the literature (including DeLancey, 1997) and in the present context – should not be confused with the (much more familiar) information structural notions of given vs. new. The latter are predominantly, if not entirely, a function of the organization and development of discourse, co-determining the topicality or focality of information and the corresponding choice of word order and intonation pattern in an utterance. In terms of this information-structural dimension, performative epistemic expressions actually appear to be used only very rarely in focus, and when they are, then apparently only under conditions of contrastivity. This suggests that they are not subject to the normal informationstructural organization of an utterance in terms of a dimension such as given vs. new. See Nuyts (2000a, 2000b), Nuyts and Vonk (1999) for empirical data and discussion.

and/or surprising to the speaker),¹⁰ and/or without having had time to share information with the interlocutor(s) (hence it is potentially new and/or surprising to those interlocutors). In opposition, intersubjectivity means that the information (and the epistemic evaluation of it) is generally known, and hence is not new (or surprising) to speaker and hearer(s). This corresponds to the fact that in corpus data the subjectivity marking mental state predicates are by far most frequent in conversational language but are nearly absent in written language,¹¹ while the intersubjectivity marking predicative adjectives hardly occur in spoken language but figure most frequently in written language (although in absolute terms they are rare even there). In conversation – especially of the type predominant in the present corpus data, viz. debates and interviews – a speaker is very often forced to formulate hypotheses and views instantaneously, in immediate response to rapidly developing and often unexpected circumstances, and very often in a ‘contrastive’ context in which speaker and hearer can be assumed not to share the same information. In writing, however, especially of the type best represented in the corpus, viz. news reporting and popular scientific reporting, the author probably is (or should be) rendering views which have been thought through well before, by himself and/or by others (scientists, politicians, etc.), and which have been gradually built up or prepared for the reader in the course of the text.

If the (inter)subjectivity dimension is seen in this perspective, then there is an obvious similarity to several dimensions which DeLancey (1997) includes in the category of mirativity, such as the marking for ‘prepared vs. unprepared minds’ in Turkish (Slobin and Aksu, 1982; Aksu-Koç and Slobin, 1986), i.e., the signaling of experience which can vs. cannot (easily) be assimilated to existing assumptions and expectations in the language user’s mind, or the marking in Lhasa Tibetan (DeLancey, 1986) and Chinese Pidgin Russian (Nichols, 1986) of the difference between new experience not yet integrated in the speaker’s mind as opposed to old experience which has already become part of his/her knowledge. (More examples in DeLancey, 1997.)

Of course, in most or all of these cases, the evidential(-like) meaning has been described in the literature in terms of the speaker’s knowledge (e.g., mirativity is usually characterized as ‘new or surprising to the speaker’). The present formulation of the dimension of (inter)subjectivity, however, assumes a crucial role for the speaker’s assumptions about the hearer’s knowledge, i.e., it (often) crucially involves the relation between speaker and hearer in the actual conversational setting (cf., e.g., antagonistic cases as discussed above). Yet, the present formulation does not stand completely alone: Choi (1995) has argued that the interpersonal dimension is also very important in Korean evidential marking of old vs. new experience. So maybe this interactive dimension is not too exceptional either, and may even be

¹⁰ So formulated, however, the category is less comparable to that expressed by Korean *-ô* as reported in Lee (1993), since the latter renders knowledge that has been part of ‘the speaker’s cognitive network’.

¹¹ They do occur in written language, but nearly always without involving speaker commitment to the evaluation. That is, they are then not used ‘performatively’ but ‘descriptively’ to report on other people’s views. See Nuyts (2000b) for details.

more widespread than has been acknowledged so far. Typological research usually works with informants and hardly ever considers real conversational or discursive data; hence it should not be surprising to find a bias in the formulation of semantic dimensions such as the present.

DeLancey (1997) argues that mirativity constitutes a system distinct from, even if intimately related to, ‘real’ evidentiality (i.e., the marking of types of information sources: inferentials, experientials, hearsay markers, etc. – cf. section 1). Although I have formulated the above argument in terms of (inter)subjectivity being an evidential category, I do not want to preclude that it might be better to consider this, too, a semantic category separate from evidentiality (strictly confining the latter to ‘types of information sources’). If so, then the apparent similarities to mirativity pointed out above suggest that it belongs to the same ‘family’ as mirativity. This discussion is for another occasion, though, as it requires a more careful look at the interactions between markings of the dimension of subjectivity and of source types.

5. A comparison with other notions of ‘subjectivity’ in the literature

Finally, to avoid potential confusion, let me briefly contrast the present notion of subjectivity with that occurring elsewhere in the literature. First of all, the present notion is not the same as that figuring in Langacker’s (1990) concept of ‘subjectification’. In the latter, subjectivity and objectivity have to do with the extent to which the ‘ground’ (i.e., the speech event, the interlocutors, and their immediate circumstances) is present in a linguistic expression. The object or event described in an expression is maximally objective and the ground maximally subjective if the ground is completely implicit in the expression. And their respective objectivity and subjectivity diminishes with an increasing ‘presence’ of the ground in the expression. As Langacker (1990: 6) states, he uses these terms “in a special, technical sense – though related, their values will not be taken as equivalent to those implied when speaking of a judgment being subjective vs. objective (i.e., ‘personal, idiosyncratic’ vs. ‘impartial, based on solid evidence’)”. How the relationship with the present notions should be seen is unclear, however. For example, in Langacker’s analysis, subjectification and grammaticalization are intimately related, and he considers the English modals to be highly subjectified (he calls them ‘grounding predications’). Yet, as argued in section 3, in the present use of the notion, the modal auxiliaries are not subjective at all. The mental state predicates, on the other hand, are highly subjective in the present analysis, yet they are clearly much less grounded, hence much less subjective than the modal auxiliaries in Langacker’s concept.

The present notion of subjectivity should also not be confused with the notion of ‘subjectification’ often found in the recent literature on grammaticalization and language change, and particularly in Traugott’s (1989, 1995, 1997; Traugott and König, 1991) and others’ writings – although the link with this notion is much more obvious. Traugott’s notion refers to the general tendency in language change (in grammaticalization processes, but also in lexical change) for forms to develop from

meanings involving the representation of the ‘object world’ (i.e., describing states of affairs) to meanings rendering the speaking subject’s attitudes or position vis-à-vis that ‘object world’.¹² One illustration of this is the common meaning development of modal auxiliaries from dynamic modal via deontic modal to epistemic modal meanings. Dynamic modality prototypically indicates an (inherent or situation-dependent) ability of the subject of the clause, i.e., it directly pertains to the internal structure of the state of affairs expressed in an utterance. Deontic modality, however, expresses the speaker’s moral assessment of the state of affairs, but also involves imposing this upon the agent/subject of the clause (as one participant in the state of affairs expressed), i.e., it is both speaker- and subject-oriented. Epistemic modality, finally, has little or nothing to do with the internal situation of the state of affairs (e.g., it is not specifically related to the subject of the clause), but involves a speaker’s evaluation of the ‘ontological’ status of the state of affairs, i.e., it is entirely speaker-oriented.

Traugott’s notion of subjectification, then, clearly relates to a theoretical concept introduced in a few functional grammar models, such as Role and Reference Grammar (Foley and Van Valin, 1984; Van Valin and LaPolla, 1997) and Functional Grammar (Hengeveld, 1989; Dik, 1997), viz. that of the hierarchical or layered representation of qualificational categories such as tense, aspect, and modality in function of their (relative) semantic scope (see also Nuyts, 1994, 2000b). Translated in terms of such models, Traugott’s subjectification essentially means semantically ‘climbing up the ladder’ of the layered system (cf. Nuyts, 1998, 2000b). The present concept of the dimension of subjectivity (regardless of how it is construed precisely, i.e., as sub/objectivity or as (inter)subjectivity) is much narrower than Traugott’s notion of subjectification, but they are related to the extent that the present dimension is situated very high in the hierarchy in a layered model of the clause. That is, the current notion of subjectivity is very high on the scale of subjectivity in Traugott’s sense.¹³

6. Conclusion

In this paper, I argued for the following three claims:

- (i) The dimension of subjectivity, as it is generally assumed to be present in epistemic modal expressions, is actually (in principle) an independent evidential-like qualificational category.

¹² Traugott’s notion is, thus, again somehow akin to Langacker’s, but the overlap is far from complete (cf. Traugott, 1995) and the precise relation is far from obvious. But see Carey (1995) for an attempt to clarify it.

¹³ The notion of subjectivity is used in yet other ways in the linguistic literature, however. See, e.g., Stein and Wright (1995) for a sample of often at first sight very similar, yet upon closer consideration subtly different uses.

- (ii) This dimension of subjectivity is expressed in some, but not in other epistemic expression types. Further, it uses linguistic resources more or less separate from (though not independent of) the epistemic forms.
- (iii) The dimension of subjectivity should not be defined in the traditional terms of the quality of the evidence for an (epistemically qualified) state of affairs, but rather in terms of the question whether the evidence (and the conclusion drawn from it) is only available to the speaker or is rather more widely known (including to the hearer).

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